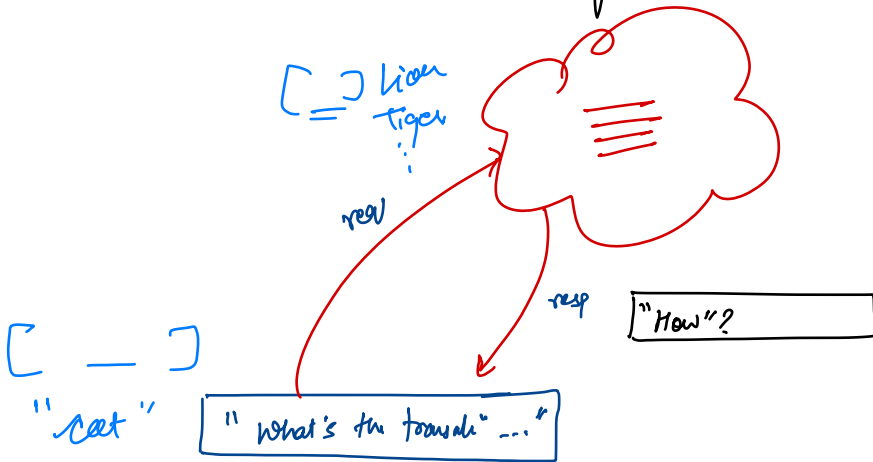


Code $\rightarrow$ Main ~~USP~~ of a developer?

Interviewer $\rightarrow$ fundamental $\rightarrow$ Architecture

Agenda

Fundamental Understanding of a Chatbot.



①

		Topic		y
1	—	Covid, Virus...		
2	—	lockdown...		

$\Downarrow$
Model

✓ One-hot-encoding

no relation $\times$ $\begin{bmatrix} \text{Covid} \\ \text{Virus} \\ \text{lockdown} \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

(UD)

Sparse

- No context
- Highly sparse

(2D)

	d_1	d_2
✓ Covid	0.7	0.5
✓ Virus	0.65	0.45
✓ lockdown	0.9	0.75
cat	7.9	7.8

⟨⟨

SVD

Singular Value Decomposition

	D_1 E_1	D_2 E_2
✓ Covid	✓ 0.5	✓ 0.1
→ Virus	✓ 0.9	✓ 0.05
lockdown	0.2	✓ 0.9
→ vaccine	0.95	0.5

Medicines Policy / govt notices

Medicines

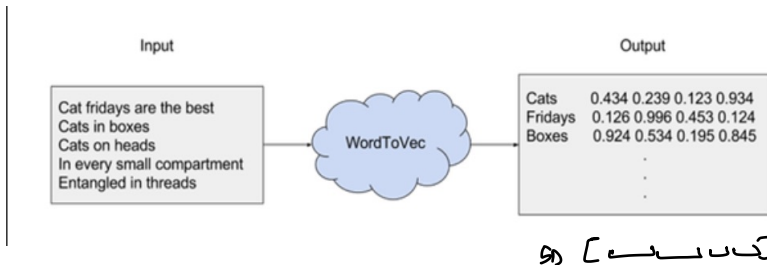
	NLP	Football
Siddhar	0.5	0.9
Snahi	0.4	0.8
Suryaj	0.9	0.2

SVD → latent pattern hidden → explain how words & content co-occur

$$X = U \Sigma V^T$$

Word 2 Vec

↳ Neural Network Method.



Architecture

Original sentence

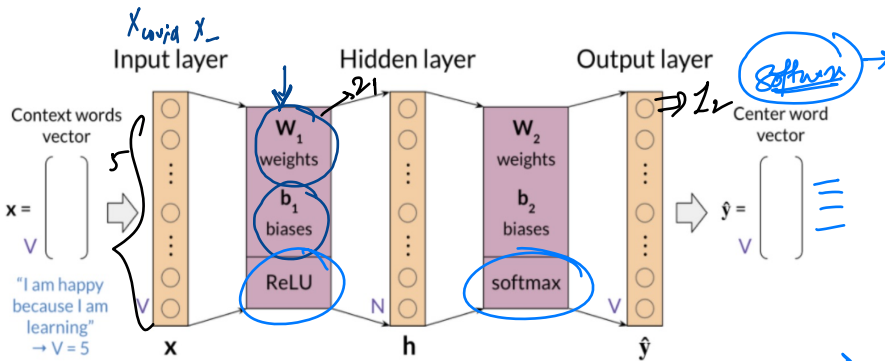
"The", "Corid", "Virus", "Spreads", "quickly"

Context "The" "Corid" "Spreads" "quickly"

Label/Target "Virus"

Continuous Bag of Words (CBOW)

Intention → predict the word from the surrounding context



X₂
Covid

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

5x1

X₂
Virus

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$Z = W^T X + b$$

$$Z = \underbrace{\begin{bmatrix} \vdots & \vdots \\ \vdots & \vdots \end{bmatrix}}_{5 \times 2} \times \underbrace{\begin{bmatrix} \vdots \\ \vdots \end{bmatrix}}_{5 \times 1} + b$$

2x5 5x1

2x1

Relu(Z)

$Z > 0$

$$Z_2 = W(Z_1) + b$$

2x5 5x1

Output $\rightarrow 5 \times 1$

$$Z_2 = [5 \times 1]$$

Softmax $\rightarrow$ Z₂ $\rightarrow$ Probability

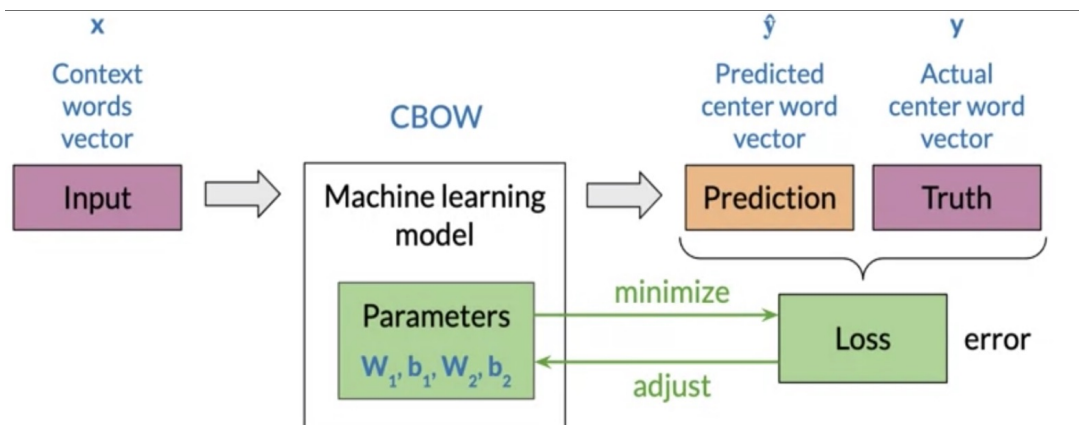
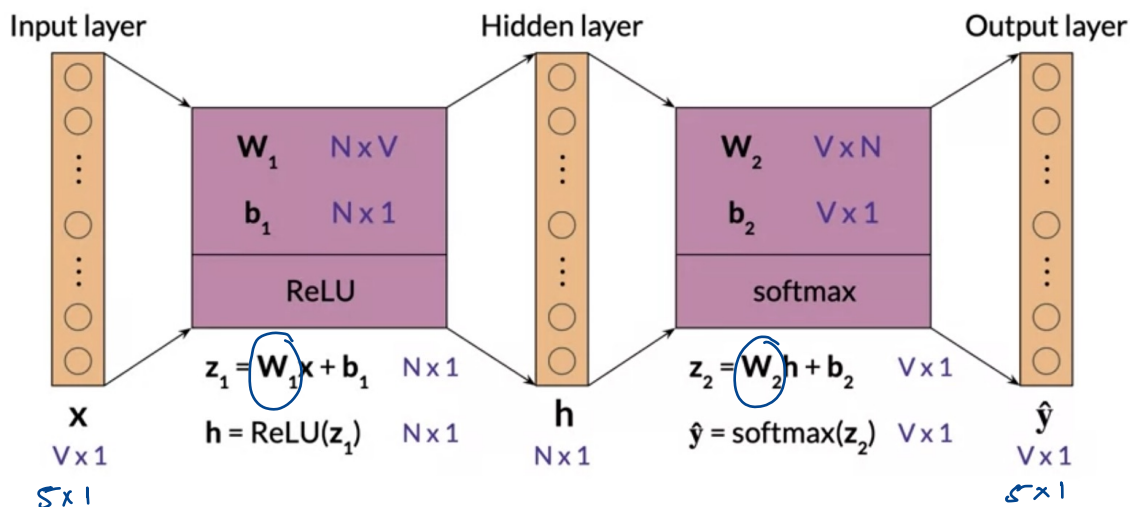
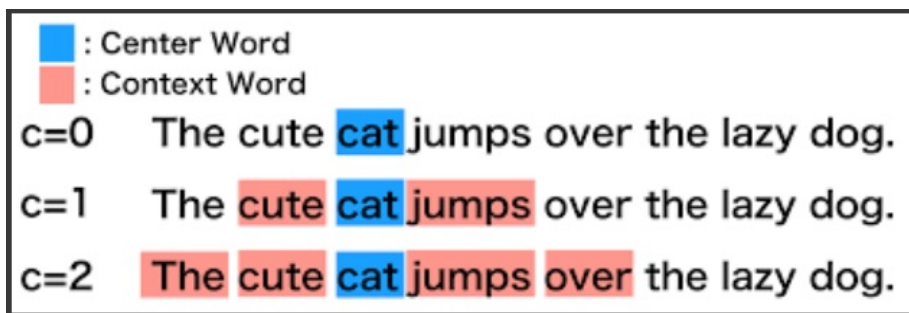
Input

Covid
Spreads
|
Solic
7

$$\begin{bmatrix} 0.65 \\ 0.75 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Virus $\rightarrow 0.9999$

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$



Skip-gram - inverse of GBow Architecture

[" " " " " " " "] $\rightarrow$ context

[] $\rightarrow$ predict

Center = "Virus"

Context = [" " " " " " " "]

4 [(center: "virus", context: "the")
 (center: "virus", context: "covid")
 (center: "virus", context: "spreads")
 (center: "virus", context: "quickly")] $\xrightarrow{P}$ higher

For each (Center, Context) pair, max the probability assigned to the true context.

Positive Samples $\rightarrow \uparrow \text{prob} \Rightarrow 1$

Negative Samples $\rightarrow \downarrow \text{prob} = 0$

softmax $\Rightarrow \frac{e^{\text{score}}}{\sum e^{\text{score}}}$

4-Classes → sklearn / statsmodel →

Langchain → API References to Query the document
↓
Retrieval Augmented
Generation